# Spatial ML - class 9 - Causal ML

Total points 100/100

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**Which question is causal rather than descriptive or predictive?** * 10/10

- ○ Which covariates are most correlated with building permit density?
- ○ Which block groups are most likely to have high permit density next year?
- ◉ Did the 606 intervention increase building permit density compared with what would have happened otherwise?
- ○ Which block groups have the highest building permit density?

**In the potential outcomes framework, what is the fundamental problem of causal inference?** *10/10

- ◉ We cannot observe both treated and untreated outcomes for the same unit.
- ○ We do not know the sample size in advance.
- ○ We cannot estimate correlations.
- ○ We do not know which model is best.

**Which variable should usually be adjusted for when estimating a total treatment effect?**

*10/10

- ( ) A post-treatment mediator
- ( ) A collider created by treatment and outcome
- (•) A pre-treatment confounder affecting both treatment and outcome
- ( ) A variable that is only strongly correlated with the outcome after treatment

**Why is a single Average Treatment Effect often not enough in applied spatial policy problems?**

*10/10

- ( ) Because causal forests do not estimate averages
- ( ) Because spatial data cannot be summarized numerically
- (•) Because treatment effects may differ across places and populations
- ( ) Because averages are never statistically valid

**What does SUTVA require?** *

10/10

- ◯ All covariates must be normally distributed
- ◯ The outcome must be measured before treatment
- ◯ Treatment must always be randomized
- ◉ No hidden versions of treatment and no interference between units

**Why is SUTVA especially difficult in spatial settings?** *

10/10

- ◯ Spatial data always have missing values
- ◉ Nearby units may be affected by the same intervention through spillovers
- ◯ Maps cannot show treatment clearly
- ◯ Spatial models cannot estimate treatment effects

**In the causal forest practical, why do untreated observations also receive predicted CATE values?** *10/10

- ◯ Because CATE is just another name for the observed outcome
- ◯ Because untreated units are removed before modelling
- ◉ Because CATE is a model-based estimate of what the treatment effect would be for each unit
- ◯ Because the model incorrectly treats all units as treated

**In the overlap diagnostic, the spatial propensity model produced much more extreme probabilities than the baseline model. What does that suggest?** *10/10

- ◯ The spatial model is misspecified
- ◉ Treated and untreated units may be less comparable once space is included
- ◯ The treatment effect is definitely zero
- ◯ There is no longer any heterogeneity

**In the comparison plot of baseline CATE vs spatial CATE, most points were below the 45-degree line. What is the best interpretation?** *10/10

- ○ The spatial model predicts larger effects for most units
- ○ The baseline and spatial models are identical
- ◉ The spatial model predicts smaller treatment effects for most units
- ○ Only treated units receive predicted effects

**In the class example, why did adding spatial features reduce the estimated treatment effect?** *10/10

- ◉ Because some of what looked like treatment effect in the non-spatial model may actually have reflected spatial structure or spatial sorting
- ○ Because spatial features automatically remove all bias
- ○ Because adding more variables always lowers treatment effects
- ○ Because causal forests cannot handle coordinates